Chapter 6

Software Quality Management

Quality

- **Quality means different to different people:**

1. Customer-Based: Fitness for use, meeting customer expectations.

2. Manufacturing-Based: Conforming to design, specifications, or requirements. Having no defects.

3. Product-Based: The product has something that other similar products do not that adds value.

4. Value-Based: The product is the best combination of price and features.

Quality

- Quality is a goal for an entire information system, not just the software portion, and it must be managed accordingly. Effective quality management assures that a process is established and followed to build quality into Information Systems. It is NOT the purpose of quality management to **"inspect in“** quality after a product is built.
- Quality software refers to a software which is reasonably bug or defect free, is delivered in time and within the specified budget, meets the requirements and/or expectations, and is maintainable.

TQM (Total Quality Management)

- TQM is the integration of all functions and processes within an organization in order to achieve continuous improvement of the quality of goods and services.
- The goal is customer satisfaction.
- TQM is the enhancement to the traditional way of doing business.
- It is a proven technique to guarantee survival in the world-class competition
- Analyzing three words (TQM), we have:

Total—*Made up of the whole*

Quality—*Degree* of *excellence a product or service provides*

Management—*Act*, *art, or manner of handling, controlling, directing, etc.*

- Therefore TQM is the art of managing the whole to achieve the excellence.

Benefit of TQM

- Greater customer loyalty
- Market share improvement
- Higher stock prices
- Reduced service calls
- Higher prices
- Greater productivity
- Improve quality
- Employment participation

Principle of TQM

- Produce quality work for the first time and every time.
- Focus on the customer
- Have a strategic approach to improvement
- Improve continuously
- Encourage mutual respect and teamwork

Six Sigma

- It is a set of techniques and tools for process improvement.
- It seeks to improve the quality of process outputs by identifying and removing the causes of defects.
- A six sigma process is one in which 99.9999966% of the products manufactured are statistically expected to be free of defects.3.4 defect per million opportunities.
- Sig sigma is a very clever way of branding and packaging many aspects of TQM (Total Quality Management).

Six Sigma

- It is a methodology for continuous improvement

- It is a set of statistical and other quality tools arranged in unique way.

- It is a methodology for creating products processes that perform at high standards.

- It is a quality philosophy and a management technique.

- It is a way of knowing where you are and where you could be

Six Sigma Objectives

✓ **Overall Business Improvement**

Six Sigma methodology focuses on business improvement. Beyond reducing the number of defects present in any given number of products.

✓ **Remedy Defects/Variability**

Any business seeking improved numbers must reduce the number of defective products or services it produces. Defective products can harm customer satisfaction levels.

Six Sigma Objectives

✓ <u>Reduce Costs</u>

Reduced costs equal increased profits. A company implementing Six Sigma principles has to look to reduce costs wherever it possibly can--without reducing quality.

✓ <u>Improve Cycle Time</u>

Any reduction in the amount of time it takes to produce a product or perform a service means money saved, both in maintenance costs and personnel wages. Additionally, customer satisfaction improves when both retailers and end users receive products sooner than expected. The company that can get a product to its customer faster may win her business.

Six Sigma Objectives

✓ **Increase Customer Satisfaction**

Customer satisfaction depends upon successful resolution of all Six Sigma's other objectives. But customer satisfaction is an objective all its own.

Label of Six Sigma

	Six Sigma Level	% Accuracy
Virtual Perfection	6	99.9997%
	5	99.98%
	4	99.4%
Good	**3.5**	**97.7%**
	3	93.3%
Improvement Needed	2	69.1%

Six Sigma Methodology

- DMAIC

  Used for project aimed at improving an existing business process.

- DMADV

  Used for projects aimed at creating new product or process design

DMAIC

- It has Five Phase

a. **Define:** define the system, the voice of customer, and their requirements, and the projects goal.

b. **Measure:** measure key aspects of the current process and collect relevant data.

c. **Analyze:** analyze the data to investigate and verify cause-effect relationships. Determine what the relationship are, and attempt to ensure that all factors have been considered.

d. **Improve:** improve the current process based upon data analysis using techniques such as design of experiments, to create a new, future state process.

e. **Control:** control the future state process to ensure that any deviations from target are corrected before they

DMADC

- It has Five Phase

a. **Define:** define design goals that are consistent with customer demands and the enterprise strategy.

b. **Measure:** measure and identify characteristics that are critical to quality, product capabilities, production process capability, and risks.

c. **Analyze:** analyze to develop and design alternatives.

d. **Design:** design an improve alternatives

e. **Verify:** verify the design, set up pilot runs, implement the production process and hand it over to process owners.

Software Quality : Defining and importance of software Quality

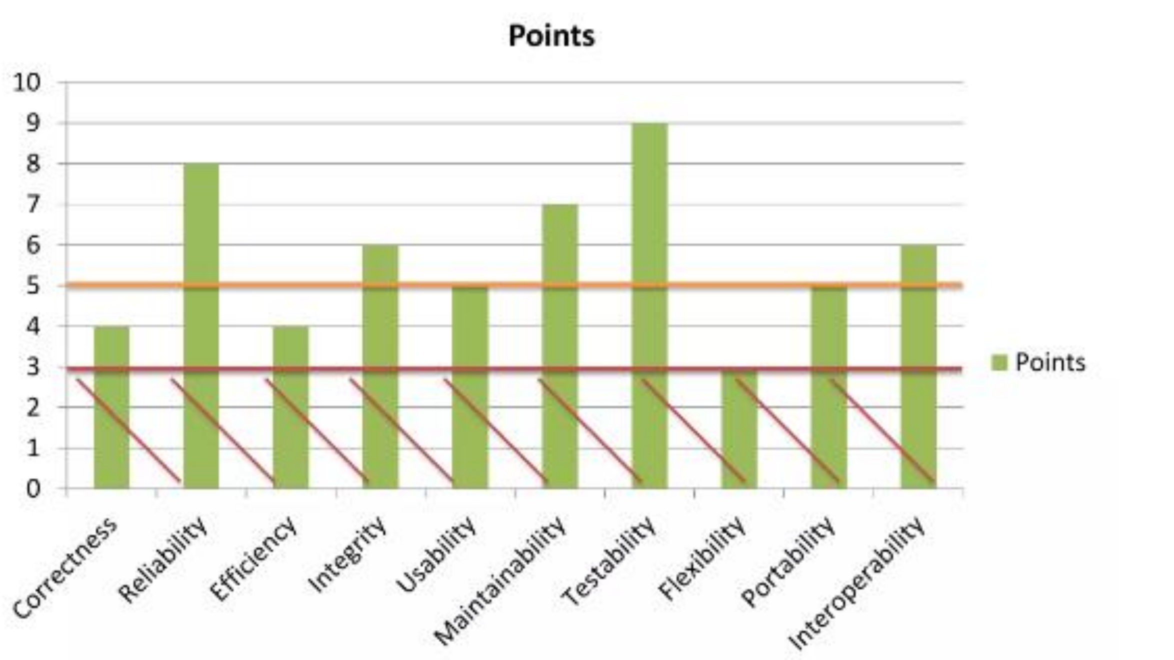
- Software quality management (SQM) is a management process that aims to develop and manage the quality of software in such a way so as to best ensure that the product meets the quality standards expected by the customer while also meeting any necessary regulatory and developer requirements.
- Software quality managers require software to be tested before it is released to the market, and they do this using a cyclical process-based quality assessment in order to reveal and fix bugs before release.
- Software quality management activities are generally split up into three core components: quality assurance, quality planning, and quality control.
- to help coordinate and align an organization's activities to uphold the customer and regulatory requirements while focusing on improving effectiveness and efficiency on a continuous basis

How to measure software quality?

1. **Correctness**: Conformance to the specifications and user requirements.
2. **Reliability**: Performing expected function with required precision.
3. **Efficiency**: Amount of resources required to execute required function.
4. **Integrity**: security features or security controls implemented in software.
5. **Usability**: Effort required to understand, learn and operate the software.

How to measure software quality?

6. **Maintainability**: Effort required to maintain the software.
7. **Testability**: Effort required to test the software.
8. **Flexibility**: Effort required to make changes to the software.
9. **Portability**: Effort required to port software from one to other platform or to configuration.
10. **Interoperability**: Effort required to couple systems with one another.



THREE CORE COMPONENTS

- **Quality Assurance**
  - Established organizational quality standards
- **Quality Planning**
  - Quality planning works at a more granular, project-based level, defining the quality attributes to be associated with the output of the project and how those attributes should be assessed
- **Quality Control**
  - The quality control team tests and reviews software at its various stages to ensure quality assurance processes and standards at both the organizational and project level are being followed

Importance of Software Quality

- **Saves time and money**
- **Strengthen security**
- **Brand reputation**
- **User satisfaction**

ISO9126

ISO 9126 standard was published in 1991 to tackle the question of the definition of software quality this 13 pages document was designed as foundation upon which further, more detailed standard could be built.

ISO 9126 identifies six software quality characteristics

- Functionality: which covers the functions that a software product provides to satisfy user needs
- Reliability: Which relates to the capability of the software to maintain its level of performance
- Usability: Efforts need to use a software
- Efficiency: physical resource used when a software is executed
- Maintainability: Effort needed to the make changes to the software
- Portability: Availability of the software to be transferred to a different environment.

ISO9126

Software quality characteristics	Sub characteristics	Description
Functionality	Suitability Accuracy Interoperability Compliance Security	
Reliability	Maturity Fault Tolerance Recoverability	Maturity → frequency of failure due to fault in software Fault Tolerance→ more identification of fault, more chance to remove them Recoverability→ control of access to the system
Usability	Understanding Ability, Learnability, operability	Understandability→ Clear quality to grasp Learnability→ Easy to learn Operability→ less time consuming

ISO9126

Software quality characteristics	Sub characteristics	Description
Efficiency	Time Behavior Resource Behavior	
Maintainability	Analyzability Changeability Stability testability	Maturity → frequency of failure due to fault in software Fault Tolerance→ more identification of fault, more chance to remove them Recoverability→ control of access to the system
Portability	Adaptability Install Ability Conformance Replace Ability	Conformance→ use of standard programming language common to many hardware / software

Place of software quality in software Planning